23CSE401 Fundamentals of AI

Prologue lab

In Fundamentals of Artificial Intelligence (AI) course, the Prolog lab serves as the practical bridge to Symbolic AI and Classical Knowledge Representation. While modern AI heavily leans toward statistical machine learning (like Python-driven Deep Learning), Prolog provides hands-on experience with how AI systems reason, infer, and map logical data.

Roles and objectives of the Prolog lab:

1. Shifting to the Declarative Programming Paradigm

Most computer science students are trained in imperative programming languages (like C++, Java, or Python), where they must tell the computer *how* to solve a problem step-by-step.

- The Prolog Shift: You learn to focus on what the problem is rather than how to compute it.
- Execution: Students supply a collection of Facts (unconditional truths) and Rules (conditional relationships), and let the language engine handle the computational execution.

2. Hands-on Knowledge Representation

A core lecture topic in any AI class is how an intelligent machine organizes information. Prolog makes abstract theories like First-Order Predicate Logic and Horn Clauses highly tangible.

- Syntax Application: You translate human language statements into computational predicate logic (e.g., mapping "Every human is mortal" into mortal(X) :- human(X).).
- Building a Knowledge Base: Practice building a family tree database to map complex, multi-generational relational facts.

3. Understanding Automated Inference and Backtracking

Instead of writing nested loops to search for answers, AI engines deduce new data from existing information.

- Resolution & Unification: Observe how Prolog automatically matches variable terms and uses backward chaining to answer user queries.
- Automatic Backtracking: When a query is initiated, Prolog uses a built-in depth-first search engine. If a path fails to find a solution, the engine automatically backtracks to explore alternate branches, giving an immediate, visual lesson in tree-based state space traversal.
